College Closures and Local Labor Markets: A Shift-Share Analysis

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Motivation: Colleges as Local Labor Market Anchors

- Colleges act as **anchor institutions**:
  - Direct local employment (faculty, staff, support workers)
  - Indirect employment via local demand (housing, services, retail)
  - Human capital accumulation and skill diversity
- Over the last decade, the U.S. has seen a notable rise in college closures, especially among **for-profit and private** institutions.
- Key labor econ question:
  - How do these closures affect **local labor markets**?
  - Are effects **heterogeneous** across places with different exposure to the education sector?

Research Question

Main Question

How do college closures affect local labor market outcomes, and how do these effects vary with local dependence on the education sector?

- Outcomes:
  - Employment rate
  - Unemployment rate
  - Labor force participation
  - Wages (median and log wages)
- Heterogeneity:
  - Baseline county-level **education employment share**
  - Quartiles of exposure vs. continuous exposure measure

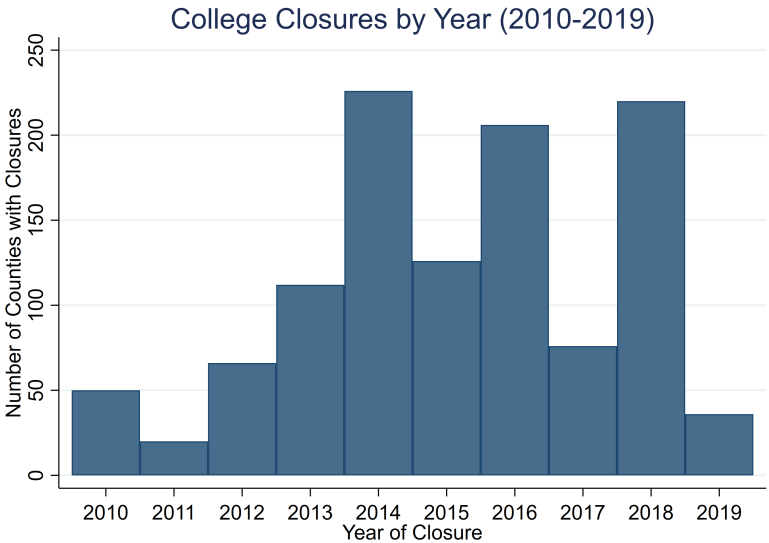
Contribution to Labor Economics

- Literature has emphasized:
  - Role of regional universities in local resilience and mobility
  - Student outcomes, debt burdens, and completion after closures
- Our contribution:
  - Treat college closures as **localized labor demand shocks**
  - Combine **difference-in-differences** with **shift-share exposure**
  - Document **heterogeneous local employment effects** by education-sector exposure
  - Provide a labor-market perspective on where closures are most damaging

Institutional Context: College Closures

- Period: 2010–2019
- Closures concentrated among:
  - For-profit colleges
  - Smaller private institutions
- Regulatory and market forces:
  - Stricter oversight of for-profit sector
  - Enrollment declines and financial distress
- Potential labor market channels:
  - Direct job loss in education sector
  - Spillovers to local non-tradable sectors
  - Changes in local skill composition and worker migration

Distribution of Closures by Year (Figure 2)



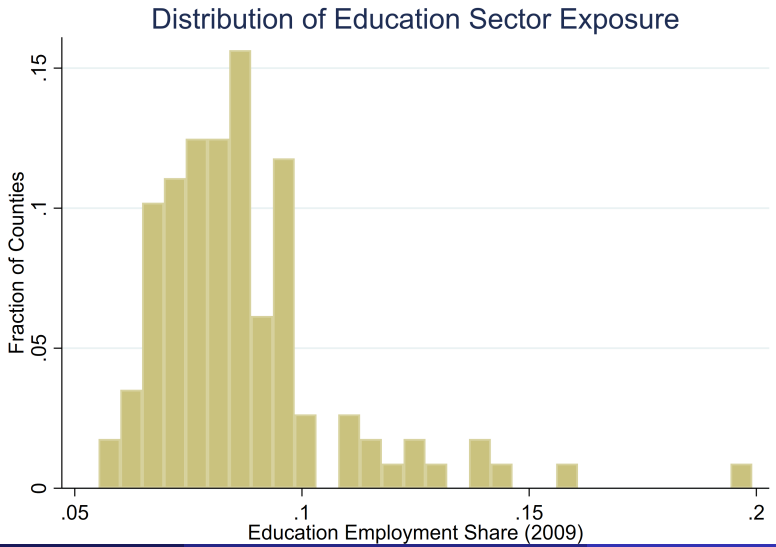
Data Sources

- **IPUMS ACS microdata**, 2009–2019
  - 108 million person-year observations
  - County-level aggregation of labor market outcomes
- **College closure data** from IPEDS
  - 213 closures
  - Assign closure timing and location to counties
- **Analytical sample**
  - 424 counties (balanced panel), 2010–2019
  - 4,056 county-year observations
  - 119 treated counties (with closures), 305 control counties (no closures)

Key Variables and Exposure Measure

- Outcomes at county-year level:
  - Employment rate, unemployment rate, labor force participation
  - Median wage, log wage
- Treatment:
  - **Treated** counties: experience at least one college closure
  - **Post** period: $t \geq 2015$ (median closure year)
- Shift-share exposure:
  - $Exposure_c$: baseline (2009) education employment share in county c
  - Average exposure $\approx 8.6\%$ among treated counties
  - Used to capture **intensity of local dependence** on education sector

Distribution of Education Sector Exposure (Figure 3)



Sample Construction

- Construct balanced panel of 424 counties over 2010–2019:
  - Require consistent ACS coverage
  - Merge in closure events by county and year
- Main analysis:
  - 119 treated counties + 305 control counties
  - 4,056 county-year observations
- Treated-only specification:
  - 119 treated counties
  - 1,138 county-year observations
  - Used mainly for robustness and comparison

Empirical Strategy: Shift-Share Difference-in-Differences

- Baseline specification:

$$Y_{ct} = \beta_0 + \beta_1 Treated_c + \beta_2 Post_t + \beta_3 (Treated_c \times Post_t) + \beta_4 (Exposure_c \times Post_t) + X_{ct}\gamma + \alpha_c + \delta_t + \varepsilon_{ct}$$

- Y_{ct}: labor market outcome in county c, year t
- $Treated_c$: indicator for ever experiencing a closure
- $Post_t$: indicator for post-2015 years
- $Exposure_c$: baseline education sector employment share
- X_{ct}: time-varying demographic controls
- α_c: county fixed effects; δ_t: year fixed effects
- Standard errors clustered at county level

Interpretation of Coefficients

- β_3: **Average treatment effect** of closures on Y_{ct} for the average county
- β_4: **Heterogeneous effect** by education exposure:
  - Effect of a 1 pp increase in baseline exposure on the post-closure change in Y_{ct}
  - Captures whether high-exposure counties are hit harder
- We also estimate:
  - Triple-difference specification with additional interactions
  - Treated-only DiD specifications

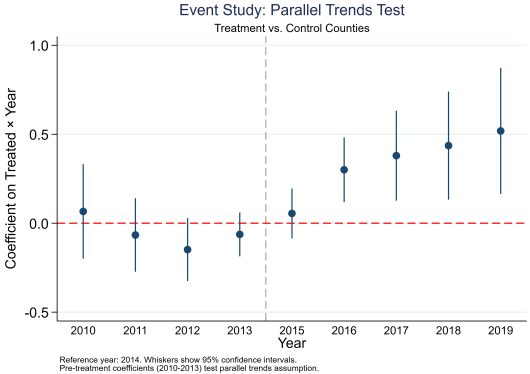
Identification and Event Study

Parallel Trends Assumption

Treated and control counties would have followed parallel trends absent closures.

Event Study Test:

- Year $\times$ Treatment interactions
- Pre-2014: insignificant
- Post-2015: positive & growing



Summary Statistics

	Pre-Period			Post-Period		
Variable	Mean	SD	Median	Mean	SD	Median
Employment Rate (%)	68.26	4.91	68.15	70.32	5.08	70.53
Unemployment Rate (%)	6.86	1.66	6.78	5.38	1.69	5.09
LFP (%)	75.12	3.97	75.03	75.70	4.16	75.97
Median Wage ($)	30,084	6,419	28,807	35,118	7,993	33,510
Education Exposure	0.086	0.020	0.082	0.088	0.022	0.084
Observations		605			533	
Counties		119			119	

- Pre vs. post periods show improving macro labor conditions
- Average education exposure $\approx 8.6\%$
- Challenge: isolate **closure effects** from aggregate trends

Main Employment Results (Table 2)

	(1) Basic DiD	(2) **With Exposure**	(3) Triple-Diff	(4) Treated Only
Treated × Post	0.444***	0.412***	0.480	
	(0.133)	(0.134)	(0.453)	
Exposure × Post		**-4.004****	-3.900*	-4.930
		(1.865)	(2.062)	(4.402)
Post	-0.247*	0.134	0.125	0.544
	(0.135)	(0.231)	(0.242)	(0.423)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	4,056	4,056	4,056	1,138

- **Column 2:** 1 pp higher exposure $\rightarrow$ 4.0 pp larger employment decline ($p = 0.032$)
- Average effect positive, but high-exposure counties are **labor-market losers**

Balance Table (Pre-Treatment, 2010)

	Treated	Control	Difference	Std. Error
Employment Rate (%)	69.78	68.97	-0.82	(0.540)
Unemployment Rate (%)	6.23	5.75	-0.48***	(0.150)
Exposure (%)	8.70	9.36	0.66**	(0.304)
% Female	50.67	50.18	-0.48***	(0.140)
% Black	15.53	10.30	-5.24***	(1.311)
% Hispanic	12.44	12.05	-0.39	(1.539)
Median Age	39.06	39.33	0.27	(0.168)
% Bachelor's+	26.53	24.25	-2.28**	(0.958)
N	119	305	424	

- Some pre-treatment differences exist
- Regressions control for these observable characteristics

Quantifying Heterogeneity

- At mean exposure ($\approx 8.7\%$):

$$\text{Net effect} \approx 0.444 + (-4.0 \times 0.087) \approx 0.10 \text{ pp}$$

- Low-exposure county (5% exposure):

$$0.444 + (-4.0 \times 0.05) \approx +0.24 \text{ pp}$$

- High-exposure county (15% exposure):

$$0.444 + (-4.0 \times 0.15) \approx -0.16 \text{ pp}$$

- Labor econ takeaway:
  - Closures act like a **localized negative labor demand shock**
  - Magnitude of the shock scales with initial concentration in education sector

Treated-Only Specifications (Table 4)

	Employment Rate		Unemployment Rate	
	(1)	(2)	(3)	(4)
Post	0.114	0.544**	-0.195**	-0.709**
	(0.125)	(0.423)	(0.091)	(0.284)
Exposure × Post		-4.930		5.891*
		(4.402)		(3.079)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,138	1,138	1,138	1,138

- Employment: -4.93 ($p = 0.265$); Unemployment: +5.89 ($p = 0.058$)
- Qualitatively similar but less precise
- Adding controls reduces SE by $\approx 58\%$

Wage Effects (Table 5)

	Median Wage ($)		Log Wage	
	(1)	(2)	(3)	(4)
Post	-53.815	109.087	-0.003	-0.020**
	(105.642)	(452.801)	(0.003)	(0.010)
Exposure × Post		-1,865		0.191*
		(4,829)		(0.099)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,138	1,138	1,138	1,138

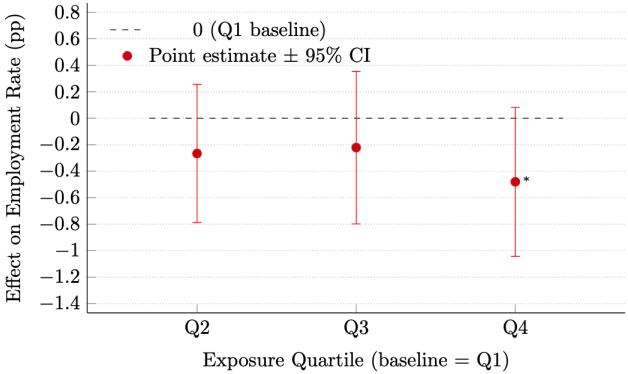
- Median wages: No significant effects
- Log wages: $+0.191$ ($p = 0.056$) suggests relative wage increases
- Possible compositional effects: lower-wage workers exit employment

Heterogeneity by Exposure Quartile (Table 6)

Dependent Variable: Employment Rate (%)		
	Coefficient	SE
Post	0.352*	(0.211)
Exposure Quartile 2 × Post	-0.266	(0.266)
Exposure Quartile 3 × Post	-0.222	(0.294)
Exposure Quartile 4 × Post	-0.480*	(0.287)
County FE	Yes	
Year FE	Yes	
Observations	1,138	

- Q4 (highest exposure, 12.4%): -0.48 pp vs. Q1 ($p = 0.097$)
- Monotonic pattern consistent with continuous exposure results

Heterogeneity Visualization (Figure 5)



Effect of College Closures on Employment by Education Exposure Quartile

Notes: Coefficients are from a DiD model interacting the post-closure indicator with exposure quartiles (Q1 omitted). Points are estimates; whiskers show 95% confidence intervals. Effects become more negative at higher exposure; Q4 shows the largest decline ($p \approx 0.10$).

Key Finding:

- Q1: 0pp (baseline)
- Q2: -0.27pp
- Q3: -0.22pp
- Q4: -0.48pp*

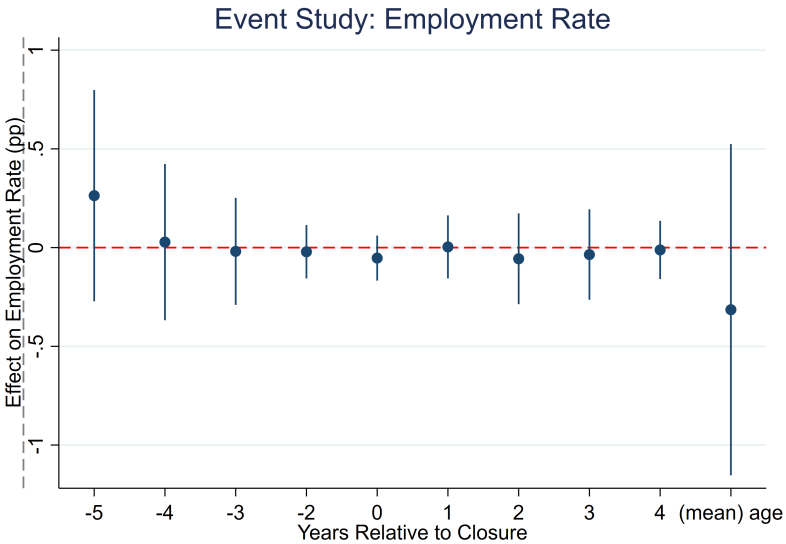
Monotonic negative pattern confirms heterogeneity

Robustness Checks (Table 7)

	(1) No Controls	(2) Basic Controls	(3) Full Controls
Post	0.462	0.492	0.445
	(0.439)	(0.440)	(0.434)
Exposure × Post	-3.559	-3.932	-3.857
	(4.590)	(4.610)	(4.564)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,138	1,138	1,138

- Coefficient stable across specifications: -3.56 to -3.93
- Not driven by omitted demographic differences

Event Study: Treated Counties Only (Figure 4)



Policy Implications

- College closures are **not just an education issue**:
  - They generate local labor demand shocks with distributional effects
- **Targeted responses**:
  - High-exposure counties may need additional support
  - Local labor market policies and safety nets should account for education-sector dependence
- Potential policy tools:
  - Worker retraining and reskilling programs
  - Local economic development and diversification strategies
  - Transitional support for displaced workers and affected communities

Limitations and Future Research

- Geographic scope:
  - 424 counties, primarily in Florida and surrounding states
  - Limits external validity but improves comparability
- Time horizon:
  - Focus on 2010–2019; long-run impacts beyond this window remain open
- Future directions:
  - National-level analysis of closures and labor markets
  - Mechanisms: worker migration, business closures, local public finances
  - Distributional impacts by worker demographic groups

Conclusion

- College closures have **heterogeneous effects** on local labor markets.
- On average, treated counties see small positive employment effects, but high-exposure counties experience **meaningful employment losses**.
- Results highlight:
  - Role of education sector as a local stabilizer
  - Importance of accounting for local industry mix in policy design
- Broad lesson for labor economics:
  - Sector-specific shocks can generate spatially uneven labor market outcomes
  - Shift-share approaches help quantify this heterogeneity

Questions?